

# GeoLabeller

## Interface Control Document

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Software version v2.0.2

Generated 2026-09-20

This document specifies the external interfaces of GeoLabeller: the project file it reads and writes, the raster products it exports, the imagery it accepts, and the coordinate conventions all of these share. It is a description of the implemented behaviour, transcribed from the source.

# 1 Scope and Conventions

GeoLabeller is a desktop application for viewing georeferenced and non-georeferenced raster imagery and recording point annotations against it. This document specifies the interfaces through which other software exchanges data with it.

## 1.1 Interfaces covered

| Interface                      | Direction        | Section |
|--------------------------------|------------------|---------|
| Project file (.geoproj / JSON) | Read and written | 3       |
| Sub-image GeoTIFF export       | Written          | 4.1     |
| Optimized GeoTIFF export       | Written          | 4.2     |
| Source imagery                 | Read             | 5       |

The HDF5 training-dataset export is specified separately and is not repeated here.

This document ships with the application: an installed copy sits beside the executable and opens from Help > ICD, so the version on hand always describes the build in front of you.

## 1.2 Notation

- Field types are JSON types: number, string, boolean, array, object.
- Pixel coordinates are (column, row) with the origin at the top-left pixel of the source image; column increases right, row increases down. Fractional values are meaningful and are not rounded on write.
- Geographic coordinates are WGS84 decimal degrees (EPSG:4326), longitude and latitude reported separately and never packed.
- Distances in metres are geodesic - measured on the WGS84 ellipsoid - not planar distances in a projected CRS. Before this they were computed on a sphere of mean radius, which reads short by 0.25% at 40 degrees latitude.

## 2 Coordinate Systems

Four coordinate systems appear across the interfaces. Only the first two are externally visible; the rest are described because they explain values that would otherwise look surprising.

| System                   | Units   | Where it appears                                                 |
|--------------------------|---------|------------------------------------------------------------------|
| WGS84 (EPSG:4326)        | degrees | Label lon/lat, waypoints, image corner coordinates               |
| Source pixel             | pixels  | Label pixel_x / pixel_y, relative to the unmodified source image |
| Web Mercator (EPSG:3857) | metres  | Internal display projection; not written to any file             |
| Scene                    | metres  | Internal canvas space; x = easting, y = -northing                |

### 2.1 Pixel coordinates are relative to the source image

Labels record the pixel they were placed on in the ORIGINAL image, at full resolution - not in the reprojected image shown on screen, and not at whatever pyramid level happened to be displayed. A label therefore addresses the same ground whatever the zoom was when it was placed, and a consumer can index the source raster directly.

### 2.2 Non-georeferenced imagery: the pixel zone

Imagery without a CRS cannot be placed geographically, so it is laid out in a reserved region of the internal scene beyond the valid extent of Web Mercator. This is an implementation detail with one external consequence: labels on such images carry pixel coordinates but no meaningful lon/lat, and those fields are written as 0.0.

| Constant             | Value      | Meaning                                              |
|----------------------|------------|------------------------------------------------------|
| PIXEL_ZONE_ORIGIN_X  | 25 000 000 | Scene X at which the non-georeferenced region begins |
| PIXEL_ZONE_SCALE     | 50         | Scene units per source pixel                         |
| PIXEL_ZONE_GROUP_GAP | 5 000      | Scene units between the columns of adjacent groups   |

### 2.3 Display resolution and memory

Imagery is displayed from pyramid overviews. The level chosen is the coarsest whose resolution is still no coarser than one screen pixel, subject to a cap on how much may be held at once. Images beyond that cap keep a coarse whole-image backdrop and read the detail for the visible area one tile at a time, so full resolution remains available at any zoom regardless of image size.

| Constant            | Value       | Meaning                                           |
|---------------------|-------------|---------------------------------------------------|
| MAX_LEVEL_PIXELS    | 150 000 000 | Largest whole-image level held in memory          |
| BACKDROP_MAX_PIXELS | 4 000 000   | Cap for the backdrop behind windowed detail tiles |
| TILE_SIZE           | 512         | Edge of a display tile, in destination pixels     |

## 3 Project File

A project is a single UTF-8 JSON document holding the class list, every image referenced, the labels on them, and any waypoints. Images are referenced by absolute path; their pixels are never copied into the project.

### 3.1 Top level

| Field             | Type            | Description                                                                                                                                                                                                                                                    |
|-------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| version           | string          | Format version of this document, currently "4.5". Governs how label coordinates are interpreted - see 3.4.                                                                                                                                                     |
| classes           | array of string | Ordered class names. A label's class must appear here.                                                                                                                                                                                                         |
| images            | array of object | One entry per referenced image; see 3.2.                                                                                                                                                                                                                       |
| waypoints         | array of object | Named geographic bookmarks; see 3.5. Absent before version 3.3.                                                                                                                                                                                                |
| descriptions      | array of string | Preset description strings for the active-description picker (applied to newly placed labels; a label's own description stays free text). Present only when non-empty; absent before version 4.2.                                                              |
| mask_names        | array of string | Preset mask names offered by the Snippet Editor when a mask is added. A name typed there joins the list. Editing it never renames a mask already painted - mask names on labels (3.3) are independent. Present only when non-empty; absent before version 4.4. |
| _next_id          | number          | Next label id to allocate. Labels ids are unique per project.                                                                                                                                                                                                  |
| _next_waypoint_id | number          | Next waypoint id to allocate.                                                                                                                                                                                                                                  |

### 3.2 Image object

| Field           | Type            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| path            | string          | Absolute path to the image file, and the key that identifies it: one entry per file. Separators are normalised on load and on every add (".." and doubled separators resolved; on Windows the separator is written as "/"), so the same raster reached through different spellings is one image rather than two. A file listed twice under two spellings has its labels merged on load. A path that does not exist is first re-tried against the project file's own folder by matching progressively longer path tails, so a project that travels with its imagery opens unchanged on another machine; anything still missing can be relocated interactively (File > Locate Missing Images). |
| name            | string          | File name without extension.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| group           | string          | Group hierarchy, '/' separated; empty when ungrouped.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| labels          | array of object | Labels on this image; see 3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| original_width  | number          | Source image width in pixels. Required to interpret label coordinates - see 3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| original_height | number          | Source image height in pixels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| reader          | object          | Extension to reader name, e.g. {"tif": "default"}.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| affine_coeffs   | array of number | Six coefficients of the source affine transform. Omitted when unknown.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| crs_epsg        | number          | EPSG code of the source CRS. Omitted for non-georeferenced imagery, and for a CRS that has no EPSG code - see crs_wkt.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| crs_wkt         | string          | WKT of the source CRS, written only when the CRS has no EPSG code (a local grid, an ESRI-defined projection, a custom datum). Exactly one of crs_epsg / crs_wkt is present on a georeferenced image. Absent before version 4.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| corners_wgs84   | object          | Convenience only: top_left / top_right / bottom_right / bottom_left, each {lat, lon}. Written when the transform and CRS are known; derived, never read back.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Field                | Type    | Description                                                                                                                                                                                                                                                                         |
|----------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| geodesic_width_m     | number  | Convenience only: ground width along the top edge, metres.                                                                                                                                                                                                                          |
| geodesic_height_m    | number  | Convenience only: ground height along the left edge, metres.                                                                                                                                                                                                                        |
| hard_negative_source | boolean | The image contains confusers but no true positives; the H5 export can opt in to sliding it whole into hard negatives. Present only when true; absent before version 3.5. Kept even while the image is not loaded, like labels.                                                      |
| location             | string  | Free-text location tag, e.g. "New York Area". Stored per image but normally applied to a whole group at once (right-click a group > Set Location...), and carried into the HDF5 export as the per-sample locations dataset. Present only when non-empty; absent before version 4.1. |

### 3.3 Label object

| Field               | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id                  | number  | Sequential id, unique within the project.                                                                                                                                                                                                                                                                                                                                                                                  |
| unique_id           | string  | UUID v4, unique to this label.                                                                                                                                                                                                                                                                                                                                                                                             |
| class_name          | string  | Must appear in the top-level classes.                                                                                                                                                                                                                                                                                                                                                                                      |
| pixel_x             | number  | Column in the source image. Stored as a FRACTION of original_width from version 2.1 - see 3.4.                                                                                                                                                                                                                                                                                                                             |
| pixel_y             | number  | Row, stored the same way.                                                                                                                                                                                                                                                                                                                                                                                                  |
| lon                 | number  | WGS84 longitude, or 0.0 for non-georeferenced imagery.                                                                                                                                                                                                                                                                                                                                                                     |
| lat                 | number  | WGS84 latitude, or 0.0.                                                                                                                                                                                                                                                                                                                                                                                                    |
| object_id           | string  | UUID v4 shared by every label of the same real-world object. Labels are linked by sharing this value. Every gt=true sample of the HDF5 training export records it in the per-sample object_ids dataset (aligned with images; "" for a hard negative, which is a view of no object), so the views of one object can be grouped, held out of a split together, or recognised as already present before appending them again. |
| length_m            | number  | Measured object length in metres. Omitted when unmeasured.                                                                                                                                                                                                                                                                                                                                                                 |
| width_m             | number  | Measured width in metres. Omitted likewise.                                                                                                                                                                                                                                                                                                                                                                                |
| description         | string  | Free text describing this labelled object. Belongs to this label alone and is never copied to labels sharing its object_id. Omitted when empty; absent before version 3.4.                                                                                                                                                                                                                                                 |
| orientation_px_rad  | number  | Unit-circle principal angle of the object's drawn orientation in UN-WARPED source-pixel space: radians in $(-\pi, \pi]$ , x along the rows (+column), y up the columns. A vector from top-right to bottom-left is about $-3\pi/4$ (-135 degrees); rightward along a row is 0; up the image is $+\pi/2$ . Omitted when never drawn; absent before 3.7.                                                                      |
| orientation_deg     | number  | The same drawn vector as a TRUE-north heading: degrees clockwise from north in $[0, 360)$ , from a WGS84 geodesic azimuth of the endpoints through the image's transform - not grid north. Omitted for non-georeferenced imagery or when never drawn. Per label even when linked: two views of one object are two measurements.                                                                                            |
| orientation_derived | boolean | Present (true) only when this orientation was NOT drawn by hand but propagated from a linked label: the group's shared true-north heading, re-derived into this image's own pixel space through its georeferencing. Cleared when the user draws over it. Absent before 3.8.                                                                                                                                                |

| Field             | Type            | Description                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| confidence        | integer         | How sure the labeller is of this label, for training: 1 (lowest) to 5 (highest), set in the Snippet Editor. Omitted when unrated. Readers must treat anything other than a whole number 1-5 as unrated - GeoLabeller loads it so. Per label, like the orientation. Absent before 4.5.                                                                                            |
| masks             | array of object | Named binary masks painted on this label's snippet in the mask editor, for measuring the object's pixel distribution against the background. Each entry is {name, x0, y0, width, height, rle}; the full encoding is specified in 3.8. From version 4.0 the window is the full source image. Present only when at least one mask exists; absent before 3.9.                       |
| group_id          | string          | Human-readable name for the linked-object group - the readable companion to object_id. The inverse contract to description: it SHALL be identical across every label sharing an object_id. Updating it anywhere updates the whole group; a file that violates the invariant is repaired on load (first non-empty name in the group wins). Omitted when empty; absent before 3.6. |
| geodesic_distance | number          | Convenience only: metres from the image's left edge at the label's row. Derived, never read back.                                                                                                                                                                                                                                                                                |

### 3.4 Label coordinate storage (important)

From version 2.1, pixel\_x and pixel\_y are stored as FRACTIONS of the image dimensions, not as absolute pixels. A consumer must multiply by the image size to recover pixel coordinates:

```
pixel_x_absolute = pixel_x * original_width
pixel_y_absolute = pixel_y * original_height
```

The conversion applies only when the document version is 2.1 or later AND original\_width and original\_height are both greater than zero. If either is zero the values are absolute pixels and must be used unchanged. Both the writer and the reader apply this rule, so a file is self-consistent; it matters when reading a project with anything other than GeoLabeller itself.

### 3.5 Waypoint object

Waypoints are named geographic bookmarks belonging to the project rather than to any image. They are navigation aids and are never exported as training data.

| Field | Type   | Description                               |
|-------|--------|-------------------------------------------|
| id    | number | Sequential id, unique within the project. |
| name  | string | Display name, e.g. "WP 1".                |
| lat   | number | WGS84 latitude in degrees.                |
| lon   | number | WGS84 longitude in degrees.               |

### 3.6 Version history

| Version | Change                                                                                                |
|---------|-------------------------------------------------------------------------------------------------------|
| 1.0     | Label-centric layout: a flat labels array, each entry naming its image. Still readable.               |
| 2.0     | Image-centric layout: images each own their labels.                                                   |
| 2.1     | Label pixel coordinates became fractions of the image dimensions (see 3.4).                           |
| 3.2     | Adds derived convenience fields: corner coordinates and geodesic extents.                             |
| 3.3     | Adds the top-level waypoints array and _next_waypoint_id. Files written before this simply have none. |
| 3.4     | Adds the optional per-label description string.                                                       |
| 3.5     | Adds the optional per-image hard_negative_source flag.                                                |

| Version | Change                                                                                                                                                                                                                                                                                                                                                                         |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.6     | Adds the shared per-group group_id string on labels.                                                                                                                                                                                                                                                                                                                           |
| 3.7     | Adds the optional orientation_px_rad and orientation_deg per label.                                                                                                                                                                                                                                                                                                            |
| 3.8     | Adds the optional orientation_derived flag for orientations propagated from a linked label.                                                                                                                                                                                                                                                                                    |
| 3.9     | Adds the optional per-label masks array of named, run-length-encoded binary snippet masks (see 3.8).                                                                                                                                                                                                                                                                           |
| 4.0     | Mask entries are written anchored to the FULL source image (x0 = y0 = 0, width/height = the image's) instead of the paint-time snippet window. The schema is unchanged and readers honor whichever window an entry records; 3.9-windowed entries are re-anchored on load. Numbered 4.0 rather than 3.10 because versions compare as strings and "3.10" would sort below "3.9". |
| 4.1     | Adds the optional per-image location string - a free-text location tag, normally applied to a whole group at once.                                                                                                                                                                                                                                                             |
| 4.2     | Adds the optional top-level descriptions array: preset strings for the active-description picker. Labels are unchanged.                                                                                                                                                                                                                                                        |
| 4.3     | Adds the optional per-image crs_wkt, carrying the source CRS of an image whose CRS has no EPSG code. Such images previously recorded an affine transform with no CRS at all.                                                                                                                                                                                                   |
| 4.4     | Adds the optional top-level mask_names array: preset names for the mask editor's picker. Masks themselves are unchanged.                                                                                                                                                                                                                                                       |
| 4.5     | Adds the optional per-label confidence rating, 1-5. Nothing else changes.                                                                                                                                                                                                                                                                                                      |

Readers should compare the version as a string and treat anything at or above a threshold as supporting that feature. Older documents load unchanged; the only migration is the 4.0 mask re-anchoring, which the reader performs structurally on load.

### 3.7 Example

A project with one image, one label and one waypoint. Note the fractional pixel coordinates: label 7 sits at absolute pixel (2048, 1024), being 0.5 x 4096 and 0.25 x 4096.

```
{
  "version": "3.4",
  "classes": ["vehicle", "building"],
  "images": [
    {
      "path": "D:/imagery/tile_014.tif",
      "name": "tile_014", "group": "survey/north",
      "original_width": 4096, "original_height": 4096,
      "reader": {"tif": "default"}, "crs_epsg": 32618,
      "labels": [
        {
          "id": 7, "class_name": "vehicle",
          "pixel_x": 0.5, "pixel_y": 0.25,
          "lon": -73.9832, "lat": 40.7536,
          "unique_id": "3b1e...", "object_id": "8f2c...",
          "length_m": 4.6, "width_m": 1.8,
          "description": "white pickup, tarpaulin over the bed"
        }
      ]
    }
  ],
  "waypoints": [{"id": 1, "name": "WP 1",
    "lat": 40.7536, "lon": -73.9832}],
  "_next_id": 8, "_next_waypoint_id": 2
}
```

### 3.8 Snippet masks and their encoding

A mask marks which pixels of a label's snippet belong to the actual object. Masks are stored per label as independent named binary layers - several may coexist and OVERLAP on one snippet - and are painted in the Snippet Editor (Labels > Snippet Editor). It keeps masks exclusive unless Allow Overlap is switched on, but that is an editing aid, not a guarantee of the format: readers must still expect overlapping masks, from projects painted with

it on and from projects older than it.

Anchoring. `x0/y0/width/height` describe the mask's window in SOURCE-image pixels. From format 4.0 the window written IS the full source image - `x0` and `y0` are 0 and `width/height` equal the image dimensions - so the decoded `rle` is co-registered with the imagery: mask pixel (`row`, `col`) is image pixel (`row`, `col`), with no anchoring arithmetic. Format 3.9 wrote the snippet crop at the moment the mask was painted instead (per the centred-window rule of 4.1). Readers SHALL honor whichever window an entry records - a mask re-projects into any other window, a different snippet size or an export crop, by pure translation and clipping - and SHOULD stream the runs rather than decode the full window, which for a 4.0 entry over survey imagery can cover billions of pixels. Only the runs bordering unpainted ground grow with the image; the run COUNT is essentially the snippet-window count.

Compression. The `rle` field run-length encodes the binary window read row-major (row 0 left to right, then row 1, and so on). Entries alternate between runs of 0s and runs of 1s, and the list ALWAYS begins with a run of 0s - a mask whose very first pixel is set therefore begins `[0, n, ...]`. The runs MUST sum to exactly `width x height`; readers SHALL reject any mask where they do not, rather than render a shifted mask. Worked example: the 4x3 window

```
0 0 1 1
0 0 1 1
0 0 0 0

encodes as rle = "2,2,2,2,4"
(2 zeros, 2 ones, 2 zeros, 2 ones, 4 zeros = 12 pixels)
```

A painted blob of a few thousand pixels typically encodes to a few hundred integers, which is why masks can live inside the project JSON and share its save/autosave/recovery behaviour rather than requiring a sidecar file.

Delivery (informative). The HDF5 training export - specified separately, see 1.1 - carries the painted masks in three aligned datasets: `mask_pixels` (`M x H x W`, uint8 binary), `mask_sample` (`M`, the images row each mask belongs to) and `mask_names` (`M`, UTF-8). Every example crop receives every mask that intersects it, re-anchored to that crop's window - so in the HDF5 there is no `x0/y0` to apply: `mask_pixels[j]` overlays `images[mask_sample[j]]` pixel for pixel as-is. Extracting them needs nothing beyond `h5py`:

```
with h5py.File(path) as f:
    for j in range(f['mask_pixels'].shape[0]):
        image = f['images'][f['mask_sample'][j]]
        mask = f['mask_pixels'][j].astype(bool)
        name = f['mask_names'][j].decode()
        object_pixels = image[mask]
```

Delivery (informative), confidence and orientation. From 2.0 the HDF5 training export also carries, aligned with images row for row: `confidence` (uint8; the label's 1-5 rating, 0 when unrated and for every hard negative), `orientation_px_rad` and `orientation_deg` (float32; the label's pixel angle and true-north heading as defined in 3.3, NaN where there is none - 0 is an angle, so it cannot mean none) and `orientation_derived` (bool; propagated from a linked label). The pixel angle holds in each crop as stored: a crop is a plain window of the source raster, which only translates it. Every crop of a label carries that label's values. Each group is written only when some exported sample has a value for it - an export with nothing to say adds no such dataset - and rows written before a group first appears read as unrated / no orientation.

## 4 Raster Exports

### 4.1 Sub-image export

Writes one GeoTIFF per label, cropped around it. The requested size is given in ground metres and converted to pixels using the true resolution at the label, so the crop covers the requested ground area whatever the CRS and whatever the pixel aspect.

| Property       | Behaviour                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------|
| Output path    | <output>/<class_name>/<object_id>_<label_id padded to 6>.tif                                                                 |
| Window         | Centred on the label's pixel, shifted (never cropped) to stay inside the image, so edge labels still yield a full-size crop. |
| Pixels         | Copied unmodified: all source bands, original data type, no rescaling.                                                       |
| Georeferencing | Source CRS preserved; transform adjusted to the window.                                                                      |
| Other metadata | nodata and per-band colour interpretation preserved.                                                                         |
| Compression    | LZW.                                                                                                                         |
| Skipped        | Labels on imagery with no CRS, labels outside the image, and windows that would be under half the requested size.            |

### 4.2 Optimized GeoTIFF export

Rewrites imagery as tiled, pyramided GeoTIFFs for fast display. Pixel values, band count, data type, CRS and transform are all preserved - only the storage layout changes.

| Property       | Behaviour                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output path    | Mirrors the group hierarchy under the output directory; keeps the source stem with a .tif extension.                                                                     |
| Layout         | Tiled, 512 x 512 blocks by default.                                                                                                                                      |
| Overviews      | Built to the requested decimation factors. Average resampling by default; paletted rasters always use nearest, so palette indices are never blended into invalid values. |
| Compression    | DEFLATE by default; may be set to another codec, none, or KEEP to retain the source's. A predictor is chosen to suit the data type.                                      |
| Large files    | BIGTIFF written when the result may exceed 4 GB.                                                                                                                         |
| Existing files | Skipped unless overwrite is requested. Output is written to a temporary file and moved into place, so an interrupted export leaves no partial .tif.                      |

## 5 Source Imagery

Imagery is read through GDAL, so any format GDAL can open may be loaded; file dialogs filter to .tif and .tiff by default, and directory import scans for those extensions.

### 5.1 Georeferenced and non-georeferenced imagery

An image with a CRS is reprojected for display and its labels carry both pixel and WGS84 coordinates. An image without one is shown at its own pixel scale in the region described in 2.2, and its labels carry pixel coordinates only, with lon and lat written as 0.0.

### 5.2 Recommendations

- Supply tiled GeoTIFFs with overviews. Without a pyramid the whole image must be decoded before anything can be shown; with one, a coarse level appears immediately and sharpens. Section 4.2 produces suitable files.
- 8-bit imagery is displayed exactly as stored. Other data types are linearly stretched for display between the 2nd and 98th percentile of a sampled read; the underlying values are never modified.
- Very large mosaics are supported: detail for the visible area is read a tile at a time, so display cost follows the viewport rather than the size of the image.
- Reads are threaded. Reprojection into the scene's projection, which is most of what displaying an image costs, and GDAL's own decoding each use about a quarter of the machine's cores, at most four. GDAL\_NUM\_THREADS is set only when the environment does not already define it, so an administrator's value wins.
- GDAL sidecar files - an external .ovr pyramid, an .aux.xml, a world file - are honoured. A folder holding none is opened without GDAL's per-open listing of the directory, which is the expensive part of an open on a network share. That is decided once per folder per session, so a sidecar added while GeoLabeller is running is picked up at the next start.

## 6 Operation

This section is included because the way a label is produced decides which fields it carries. A consumer reading a project file will meet labels that share an `object_id`, labels with measurements and labels whose lon/lat are zero, and 6.3 explains where each comes from.

### 6.1 Interaction modes

The canvas is always in exactly one mode. Modes differ in what a left click does and how the view is navigated; none of them change how a label is stored.

| Mode       | Left click          | Purpose                                                                                                                                                                                             |
|------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pan        | Drags the view      | Navigating. Right click also offers waypoint and layer actions.                                                                                                                                     |
| Label      | Places a label      | Ordinary labelling of whatever is on screen.                                                                                                                                                        |
| Cycle      | Places a label      | Steps through a group one image at a time; Space advances, Ctrl+Space goes back. At either end the image stays showing and a short notice says so; C there starts the group again.                  |
| View Cycle | Places a label      | As Cycle, but over the images currently in view.                                                                                                                                                    |
| Waterfall  | Places a label      | Stacks a bottom-level group vertically as raw pixels and glides through it; every label in the group stays visible. Leaving the mode returns the view to the geography of the image being examined. |
| Ruler      | Drags a measurement | Reading ground distances. Records nothing in the project.                                                                                                                                           |

Two overlays work on top of whichever labelling mode is active, rather than replacing it:

| Overlay                     | Effect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chain Link                  | Clicking labels one after another links them into a single object; each click writes the shared <code>object_id</code> immediately. Right-clicking a label also offers Link by Box: drag a desktop-style box and every label inside is linked to that label's object. Linked groups always wear a coloured halo ring - one colour per group, distinct between groups, updated the moment links change.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Measure                     | Draws a length and then a width line on a label, writing <code>length_m</code> and <code>width_m</code> . Needs source georeferencing, since metres are undefined without a CRS - but works in waterfall mode too: the drawn line is mapped through source pixels back to the ground, so stacked and geographic measurements agree.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Description                 | Right click a label to write a free-text note on it, shown as the marker's tooltip. Stays with that label when it is linked. A project may also carry preset descriptions (3.1); the active preset - toolbar picker, or Shift+1-9, pressing it again or Shift+0 for none - is applied to every newly placed label, and editing the presets never changes labels already written.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Locate missing images       | File menu. Rebinds a shared project's images to their paths on this machine: search a folder (filename matches verified against each image's recorded size and CRS) or locate one file and apply its old-versus-new path prefix to the rest. Applied paths persist when the project is saved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Recent projects and folders | File > Open Recent lists the projects opened or saved lately, newest first, and opens one directly; an entry whose file has moved says so and drops off the list. Each kind of file dialog also reopens where it was last used - imagery, project files and exports are remembered apart. Both are per user (they are how somebody works, not what the work is), so neither travels with a project file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Import ground truth         | File menu. Adds the labels from a GT file (Export > Ground Truth, or any project file) made on another machine to the current project, whose images were loaded here - typically with Add Directory. Each GT image is matched to a loaded image by filename (case-insensitive, read by the writing OS's path rules), verified against the size and CRS the GT recorded; several fits go to the longest shared folder tail, and a tie matches nothing. Labels keep their pixel positions, fields, <code>unique_id</code> and <code>object_id</code> , and take new ids; one whose <code>unique_id</code> is already in the project is skipped, so importing again adds only what is new. Their classes and the GT's presets are added; a hard-negative flag or location tag is taken only where the local image has none; waypoints are not imported. A preview lists each GT image's outcome before anything changes, and labels on images that cannot be matched are left out. |

| Overlay                          | Effect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Snippet Editor                   | Labels > Snippet Editor... is one window for everything recorded per snippet; in 2.0 it replaced the separate Orientation Editor and Mask Editor windows. A list of the project's snippets - one class or all, in class, image and label order, with a Show filter offering 'needs' and 'done' for each section - feeds two views: Single, one snippet at a time and zoomable, with the Orient and Paint masks tools; and Grid, a page of snippets at once, where double-clicking one opens it in the Single view. A column of sections - Orientation, Masks, Confidence - each switches on and off, taking its panel, filter pair, overlay, tool and keys with it. Everything is written to the project as it is done; the save bar says whether it is on disk yet (Ctrl+S saves).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Orient                           | In the Snippet Editor's Grid view, or with the Orient tool in its Single view, dragging tail-to-nose across a snippet stores orientation_px_rad and, for georeferenced imagery, orientation_deg; right-click clears it. Snippets are raw source pixels with the export's framing, so the drawn vector is a source-pixel vector. Oriented snippets get a coloured border: amber for hand-drawn, violet for propagated. With the 'apply heading to linked labels' toggle on, drawing on a linked label gives the group's other labels the same true-north heading, each with a pixel angle derived through its own image (marked orientation_derived until drawn over by hand).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Mask                             | In the Snippet Editor's Single view, the Paint masks tool paints named binary masks on a label's snippet (left-drag paints, right-drag erases). The mask to paint is chosen by an active mask name, picked from or typed into the project's mask_names list (3.1); Add Mask creates it and selects it in one click. Allow Overlap (off by default, remembered per user) decides whether a stroke may cover another mask's pixels; while it is off, other masks also act as edges for Fill Enclosed. The snippet size is selectable; masks are stored anchored to the full source image, so resizing changes only the window being painted - existing masks are never moved or cropped by it. The wheel zooms to the cursor (0.25x-32x, pixels stay crisp) and Shift+drag pans, so single-pixel edits are practical. Space / Ctrl+Space step through the snippet list, whose Show filter offers the snippets that need masks or already have them. Fill Enclosed floods the interior of a drawn outline, and refuses when the outline has a gap (nothing enclosed). Brightness and contrast sliders change the display only. A live readout compares the object's RAW per-band mean/std against the background. Stored per 3.8; carried into the HDF5 export automatically. |
| Confidence                       | In the Snippet Editor's Confidence section, the buttons - or keys 1-5 in the Single view - rate a snippet from 1 (lowest) to 5 (highest); 0 clears the rating. Stored as the label's confidence (3.3), omitted when unrated, and exported to HDF5 as the per-sample confidence dataset, 0 for unrated labels and for negatives (3.8).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Group location                   | Right-click a group > Set Location... tags every image in the group (sub-groups included) with a free-text location such as "New York Area". The tag is stored per image (3.2), so it survives regrouping, and every HDF5 sample cut from a tagged image records it in the per-sample locations dataset (aligned with images; absent when nothing is tagged). An empty entry clears the tag.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Remove a layer or group          | Right click in the layer panel. This DELETES the images and their labels from the project, not just from the view: the confirmation states how many of each, and a single unlabelled image is removed without asking. Hiding without deleting is the visibility checkbox.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Flag hard negatives              | Right click an image to mark it as a hard-negative source: it holds confusers but no true positives. Flagged images collect in the Hard Negatives panel, and the H5 export can take its hard negatives from them alone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Export one object                | Right click a label on the canvas, or an "Object: ..." row in the labelled panel, > Export Object Snippets... The HDF5 export opens scoped to that linked object: its true-positive snippets from every image it appears in, or only from the layers toggled on. Instances keep their own class names, so a linked object whose views disagree exports as it was labelled - the dialog shows the breakdown first.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Choosing what an export contains | The HDF5 dialog asks two independent questions: which images contribute gt=true examples (a selected object, the layers toggled on, all loaded layers, or nothing) and which get the hard-negative window slid over them (nothing, the same images as the examples, the layers toggled on, flagged sources only, or all loaded layers). Labelled ground is never written as a hard negative whichever images are slid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## 6.2 How a label's fields are produced

Every field in 3.3 is written for every label; these are the ones whose value depends on how or where the label was made.

| Field             | Produced by                                            | Consumer note                                                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pixel_x, pixel_y  | Any left click in a labelling mode                     | Always the source image's own pixels at full resolution, whatever the zoom or pyramid level on screen; a click that falls outside them - the transparent margin a reprojected raster leaves inside its bounding box - places no label at all. Stored as fractions - see 3.4. |
| lon, lat          | Derived from the clicked pixel through the image's CRS | Zero when the image has no CRS. Check crs_epsg (or crs_wkt) on the image before trusting these.                                                                                                                                                                              |
| object_id         | Shared when labels are linked                          | Unique per label until then. Labels of one real object are found by grouping on this value, not by proximity.                                                                                                                                                                |
| length_m, width_m | Measure overlay                                        | Absent when never measured. Geodesic metres, not planar. Copied to every label sharing the object_id when the linked-measurement option is on.                                                                                                                               |
| description       | Right click a label, Description...                    | Free text, absent when empty. Per label even when labels are linked: the same object in two images is described separately.                                                                                                                                                  |
| group_id          | Right click a label, Group ID...                       | One value for the whole linked group, always: setting it on any member applies it to every label sharing the object_id. A label that is unlinked loses it; a label joining a named group adopts the name (the first label's name wins a merge conflict).                     |
| class_name        | The class selected when the click was made             | Always one of the top-level classes.                                                                                                                                                                                                                                         |

### 6.3 Keyboard and mouse reference

Generated from the application's own binding table, so it matches the reference shown by F1 exactly.

| File Operations |                              |
|-----------------|------------------------------|
| Ctrl+N          | New Project                  |
| Ctrl+Shift+P    | Open Project                 |
| Ctrl+S          | Save Project                 |
| Ctrl+Shift+S    | Save Project As              |
| Ctrl+O          | Add Image (GeoTIFF + custom) |
| Ctrl+Shift+O    | Add Directory                |
| Ctrl+Q          | Exit                         |

| Navigation   |                               |
|--------------|-------------------------------|
| Mouse Wheel  | Zoom in/out                   |
| Click + Drag | Pan (in Pan mode)             |
| Right-click  | Context menu                  |
| Ctrl+G       | Go to a latitude/longitude    |
| Ctrl+Shift+W | Add waypoint by coordinates   |
| Ctrl+B       | Show/hide the snippet sidebar |

| Mode Switching |                                         |
|----------------|-----------------------------------------|
| P              | Pan mode                                |
| L              | Label mode                              |
| C              | Cycle mode (group-based)                |
| V              | View Cycle (layers in view)             |
| W              | Waterfall mode (stack group vertically) |

| Mode Switching                   |                                            |
|----------------------------------|--------------------------------------------|
| R                                | Ruler mode                                 |
| Labeling (Label & Cycle modes)   |                                            |
| Left-click                       | Place label                                |
| Right-click label                | Label options (remove, link, measure)      |
| Right-click + drag               | Pan the view                               |
| Ctrl+Left-click label            | Label options (Cycle modes)                |
| 1-9                              | Quick-switch to class 1-9                  |
| Shift+1-9                        | Description preset 1-9; again for none     |
| Shift+0                          | No description (new labels get none)       |
| Escape                           | Cancel link mode                           |
| Chain Linking                    |                                            |
| K                                | Toggle chain-link (any labeling mode)      |
| Left-click label                 | Add to chain (first click anchors)         |
| N                                | Finish this chain, start a new one         |
| Escape                           | Exit chain-link (links are kept)           |
| Cycle Modes (Cycle / View Cycle) |                                            |
| Space                            | Next layer (unchecks current)              |
| Ctrl+Space                       | Go back to previous layer                  |
| Left-click                       | Place label                                |
| Right-click label                | Label options (remove, link, measure)      |
| Right-click + drag               | Pan around                                 |
| Mouse wheel                      | Zoom in/out                                |
| Waterfall Mode                   |                                            |
| Hold Space                       | Glide up through the image stack           |
| Hold Ctrl+Space                  | Glide down (starts at the bottom)          |
| + / -                            | Glide speed up / down                      |
| Left-click                       | Place label (group labels stay visible)    |
| Right-click label                | Label options (remove, link, measure)      |
| Right-click + drag               | Pan within the stack                       |
| Mouse wheel                      | Zoom in/out                                |
| Measuring                        |                                            |
| Shift + Drag                     | Measure distance, without changing mode    |
| M                                | Measure label under cursor (length, width) |
| Escape                           | Clear the measurement or the go-to marker  |
| Waypoints                        |                                            |
| Right-click map                  | Add waypoint here (Pan mode, geo)          |
| Ctrl+Shift+W                     | Add waypoint by coordinate                 |
| Right-click marker               | Go to / Rename / Remove                    |
| Double-click in panel            | Fly to that waypoint                       |

|                             |                                              |
|-----------------------------|----------------------------------------------|
| <b>Waypoints</b>            |                                              |
| Show on map                 | Hide or show every waypoint marker           |
| <b>Layer Panel</b>          |                                              |
| Checkbox                    | Toggle layer/group visibility                |
| Right-click group           | Select/Unselect, Expand/Collapse all         |
| Right-click layer           | Zoom to layer, Remove (deletes from project) |
| Drag & Drop                 | Reorder layers/groups                        |
| <b>Labeled Images Panel</b> |                                              |
| Checkbox                    | Toggle image visibility (synced)             |
| Right-click label           | Zoom to label or layer                       |
| Right-click group           | Select/Unselect all in group                 |
| <b>Snippet Editor</b>       |                                              |
| Space                       | Next snippet (Single view)                   |
| Ctrl+Space                  | Previous snippet (Single view)               |
| 1-5                         | Rate confidence, 1 lowest (Single view)      |
| 0                           | Clear the confidence rating                  |
| Left-drag                   | Paint the mask, or draw a heading (Orient)   |
| Right-drag                  | Erase the mask (Paint)                       |
| Right-click                 | Clear the heading (Orient, or a grid cell)   |
| Shift+drag                  | Pan the zoomed snippet                       |
| Mouse wheel                 | Zoom to the cursor                           |
| Double-click                | Open a grid cell in the Single view          |
| Ctrl+S                      | Save Project                                 |
| <b>Help</b>                 |                                              |
| F1                          | Show this help                               |